HEP Applications

Computing for High-Energy Physics

8-Week Self-Paced Course Syllabus and Lesson Plan

Course Description

This self-paced course covers the modern computing landscape for high-energy physics research at RHIC and the LHC. Students progress from C++ fundamentals and the ROOT data analysis framework through machine learning, deep learning, large language models, AI agents, edge computing for trigger systems, and Monte Carlo simulation with Geant4. The course culminates in a capstone project that integrates skills across the full sPHENIX analysis chain, from event generation with PYTHIA/HIJING through detector simulation, reconstruction, and ML-based physics analysis.

Prerequisites

Intermediate familiarity with C++ programming and basic Linux command-line usage. Prior exposure to physics data analysis is helpful but not required. All software tools will be introduced from the ground up.

Learning Outcomes

1. Write modern C++ code (C++11 through C++20) for efficient HEP data analysis and simulation

2. Use the ROOT framework (TFile, TTree, TH1, RDataFrame) and Python tools (uproot, awkward-array) for data handling

3. Build, train, and evaluate machine learning models with scikit-learn and PyTorch for particle classification

4. Apply advanced deep learning architectures (GNNs, autoencoders, normalizing flows, PINNs) to HEP problems

5. Leverage large language models for code generation, build RAG systems, and create AI agent workflows

6. Deploy ML models on FPGAs using hls4ml and optimize inference with ONNX Runtime and TensorRT

7. Construct Geant4 detector simulations with PYTHIA8 and HIJING event generators for sPHENIX physics

Weekly Schedule

Week	Topic	Modules	Readings	Key Concepts
Week 1	C++ Review + ROOT	Modules 0, 0b	Readings 00, 00b	Variables, types, OOP, STL, TFile, TTree, histograms, RDataFrame, PyROOT, uproot
Week 2	Modern C++	Module 1	Reading 01	auto, smart pointers, RAII, move semantics, templates, constexpr, C++17/20, concurrency
Week 3	ML/AI First Steps	Module 2	Readings 02, 02b	NumPy, scikit-learn, neural networks from scratch, PyTorch, CNNs, evaluation metrics
Week 4	Deep Learning for HEP	Module 3	(Week 3 readings)	GNNs, autoencoders, PINNs, normalizing flows, transformers, uncertainty quantification
Week 5	LLMs + Agents	Modules 4, 5	Reading 03	Transformer architecture, prompt engineering, fine-tuning/LoRA, RAG, Claude API, MCP, LangChain
Week 6	Edge + Frameworks	Module 6	Reading 04	hls4ml, quantization, FPGAs, JAX, ONNX, TensorRT, MLOps with MLflow
Week 7	Geant4 Part A	Module G4A	Reading 05a	Geant4 architecture, geometry, materials, physics lists, primary generators, visualization
Week 8	Geant4 B + Capstone	Modules G4B, 7	Reading 05b	Sensitive detectors, PYTHIA8, HIJING, Fun4All, sPHENIX subsystems, full analysis pipeline

Reading Assignments

PDF Filename	Title	Week
Reading-00	C++ Review Fundamentals	Week 1
Reading-00b	ROOT Framework for HEP	Week 1
Reading-01	Modern C++ for HEP	Week 2
Reading-02	ML Foundations for Physics	Week 3
Reading-02b	AI/ML First Steps	Week 3
Reading-03	LLMs, Agents and Vibe Coding	Week 5
Reading-04	Edge Computing and Frameworks	Week 6
Reading-05a	Geant4 Basics and Geometry	Week 7
Reading-05b	Geant4 Physics and sPHENIX	Week 8

Assessment Methods

Quizzes: Each module includes a 5-question multiple-choice quiz covering key concepts. Quizzes are self-graded within the interactive course application.

Flashcards: Each module provides 6 flashcards for spaced-repetition review of terminology, commands, and core concepts.

Hands-On Projects: Every module culminates in a terminal-based project with real commands: compiling C++ code with g++, running ROOT macros, training PyTorch models, building LangChain agents, synthesizing FPGA firmware, and running Geant4 simulations.

Capstone Project (Week 8): A comprehensive project integrating skills from all modules into a complete sPHENIX analysis pipeline covering event generation, detector simulation, reconstruction, and ML-based jet classification.

Course Information

Format: Self-paced, interactive web application with offline PDF readings

Duration: 8 weeks (recommended), flexible pacing

Time Commitment: 10-15 hours per week

Prerequisites: Intermediate C++, basic Linux command line

Software: See Setup and Installation Guide PDF